

Customer_Rental Table

Customer_No	CName	Property_No	PAddress	RentStart	RentFinish	Rent	Owner_No	OName
CR76	John Kay	PG4	6 Lawrence St, Glasgow	01/07/1994	31/08/1996	350	CO40	Tina Murphy
		PG16	5 Novar Dr, Glasgow	01/09/1996	01/09/1998	450	CO93	Tony Shaw
CR56	Aline Stewart	PG4	6 Lawrence St, Glasgow	01/09/1992	10/06/1994	350	CO40	Tina Murphy
		PG36	2 Manor Rd, Glasgow	10/10/1994	01/12/1995	375	CO93	Tony Shaw
		PG16	5 Novar Dr, Glasgow	01/01/1996	10/08/1996	450	CO93	Tony Shaw

Important Note :

 Highlighted Text → PK

 Underlined Text → FK

0 Normal Form :

Customer (**Customer_No** , Cname , Property_No , Paddress ,
RentStart , RentFinish , Rent , Owner_No , Oname)

1. 1NF — First Normal Form

- A relation must contain **no Multivalued Attributes**.
- No **Repeating Groups**.
- No **Composite Attributes**.
- To achieve 1NF:
 - **Repeating Group:** Move it to a new table + add the original PK as a FK.
 - **Multi-valued Attribute:** Move it to a new table + add the original PK as a FK.
 - **Composite Attribute:** Split its subparts into separate columns.

Customer (**Customer_No** , Cname)

Rental (**Customer_No , Property_No** , Paddress , RentStart , RentFinish ,
Rent , Owner_No , Oname)

2. 2NF — Second Normal Form

- Must be in **1NF**.
- No **Partial Dependency**.
- Every Non-Key Attribute must depend on the **whole PK**.
- To achieve 2NF:
 - Move the partially dependent Non-Key Attributes to a new table.
 - Add the **part of the PK they depend on** as the new table's PK/FK.

Customer (**Customer_No** , Cname)

Rental (**Customer_No , Property_No** , RentStart , RentFinish , Rent)

Property (**Property_No** , Paddress , Owner_No , Oname)

3. 3NF — Third Normal Form

Must be in 2NF.

- No **Transitive Dependency**.
- A Non-Key Attribute must not depend on another **Non-Key Attribute**.
- To achieve 3NF:
 - Move the dependent Non-Key Attributes to a new table.
 - Keep the Non-Key Attribute they depend on in the new table.
 - Keep that determining Non-Key Attribute in the original table as well.

Customer (**Customer_No** , Cname)

Rental (Customer No , Property_No , RentStart , RentFinish , Rent)

Property (Property No , Paddress , Owner No)

Owner (**Owner_No** , Oname)